Enhancing Detection of Left Ventricular Abnormalities on Chest Radiographs Using Modern Deep Learning Architectures

Cristina-Andreea Szabo
Faculty of Automatic Control and Computer Science
University Politehnica of Bucharest
Bucharest, Romania
cristina.szabo@stud.acs.upb.ro

Luca Botez
Faculty of Automatic Control and Computer Science
University Politehnica of Bucharest
Bucharest, Romania
luca.botez@stud.acs.upb.ro

I. Dataset description

The CheXchoNet dataset [1] consists of 71,589 unique chest X-rays from 24,689 different patients alongside key echocardiography measurements indicative of left ventricular hypertrophy and dilated left ventricle (ivsd, which measures the septal wall thickness; lvpwd, the thickness of the left walls of the heart; and lvidd, the overall size of the left ventricle). The dataset also includes temporal information for each radiograph (cxr_time_offset, cxr_year), patient demographic attributes (age, sex), and additional metadata (heart_transplant, lung_transplant, pacemaker_or_icd). This comprehensive multimodal resource enables the development and evaluation of advanced models for detecting left ventricular abnormalities.

All samples have been collected from the Columbia University Irving Medical Center, including patients who had both an CXR and echocardiogram, providing clinically confirmed cardiac diagnoses, conducted from January 2013 to August 2018.

The dataset [1] exhibits an uneven class distribution, containing 86.2% Healthy samples, 7.7% SLVH, 5.1% DLV, and 0.9% patients with both diseases, a key aspect to consider when developing predictive models.

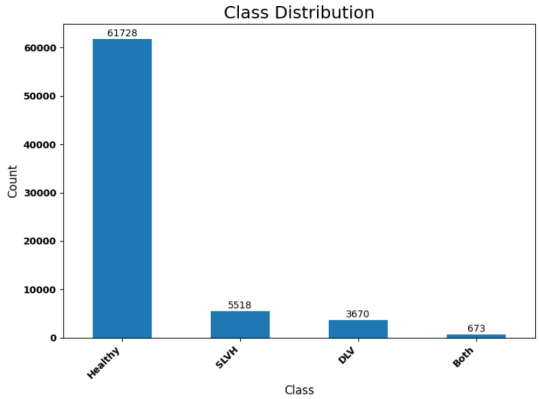

Fig. 1. Class distribution across patient groups.

Demographically, the dataset [1] covers a broad range of age groups, with the highest concentration occurring between 50 and 80 years. The gender classes are balanced, eliminating the risk of gender-related bias in subsequent model evaluation.

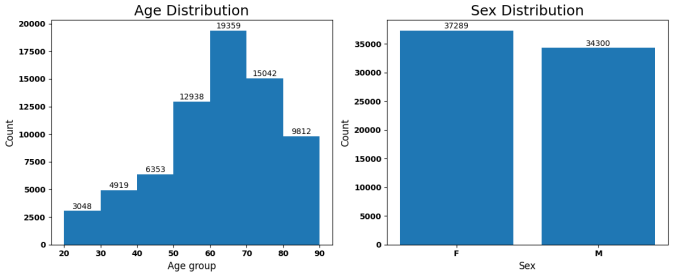

Fig. 2. Age and Gender distribution across all patients.

II. Data preprocessing

Several standard preprocessing procedures steps have already been applied to the CheXchoNet data collection [1], in order to make the data tractable and easy to use for downstream application, such as downsampling to 224-by-224 pixel images using bicubic interpolation and contrast-limited adaptive histogram equalization.

In addition, the demographic variables were preprocessed by standardizing age using the mean and standard deviation computed from the training set and by one-hot encoding the sex attribute. To create a reliable baseline backed up by scientific research, each step of the Bhave et al. paper [2] has been followed, including the normalization of the dataset with the overall mean and variance and the addition of gaussian noise using normal distribution with mean 0 and variance 0.05, both aiming to improve generalization performance.

For the purposes of this study, a balanced subset consisting of 5,000 healthy cases and an equal number of SLVH and DLV samples, alongside all the available combined-pathology subjects has been extracted, aiming at fair and unbiased model training and evaluation across all diagnostic categories.

Following the construction of the balanced subset, the data were partitioned into training, validation, and test sets using a patient-level split to prevent information leakage across partitions. Specifically, 90% of the samples were allocated to the training set and 10% to the test set, out of which 5% form the validation set, ensuring that no images from the same patient appear in more than one subset.

III. Baseline model

The architecture applied uses DenseNet121 [3] as the backbone, having the last layer responsible of classifying the input replaced with a feed forward layer that generates mean and variance parameters for three independent univariate normal distributions, one for each echocardiographic measurements. Subsequently, these distributions are converted to probabilities for each of the three binary labels indicating structural abnormalities.

The DenseNet model [3], which consists of 4 dense blocks and 3 transition layers, is pretrained on ImageNet, and produces a 1024-dimensional representation via global average pooling; this image feature vector is then concatenated with the standardized age and one-hot encoded sex inputs, and passed through a fully connected prediction layer. Model optimization follows the approach described by Bhave et al. [2], using the Adam optimizer [4] for stable gradient-based learning and a Gaussian negative log-likelihood [5] loss to supervise the estimation of the mean and variance parameters for each echocardiographic measurement.

To convert the model's regression outputs into disease probabilities, the same probabilistic formulas used in the base study are applied. For each echocardiographic measurement, IVSd, LVPWd, and LVIDd, the network predicts a mean value and a log-variance, defining a Gaussian distribution that reflects both the estimated value and its uncertainty. Using these distributions, we compute the probability that each measurement exceeds the clinically established thresholds: IVSd > 1.5 cm, LVPWd > 1.5 cm, and LVIDd > 5.9 cm, associated with left ventricular hypertrophy or dilation. The probability of SLVH is obtained by combining the IVSd and LVPWd probabilities using the standard formula for the union of independent events, while DLV is derived directly from the probability that LVIDd exceeds its threshold. A composite probability is then formed by combining SLVH and DLV in the same manner.

IV. Evaluation metrics

The model shows a clear upward trend in AUC for all three tasks, SLVH, DLV, and Composite, indicating that it successfully learns meaningful features from chest radiographs. The Composite prediction performs best overall, while SLVH and DLV show more modest improvements. Training loss decreases steadily, while validation loss remains more variable, likely due to label noise in echocardiographic measurements and the limited size of the reduced subset used for training. These patterns are reflected in the final test results, where the model reaches an AUC of approximately 0.73 for SLVH, approximately 0.75 for DLV, and approximately 0.79 for the composite outcome, with a test loss of approximately 0.41. Overall, the model generalizes reasonably well given the limited dataset and the complexity of the targets. Importantly, these values are close to the performance reported in the original article, which achieved composite AUCs around 0.80–0.83 on the full dataset. Considering that our model is trained on a smaller subset, the results obtained indicate behavior consistent with the findings of the original study.

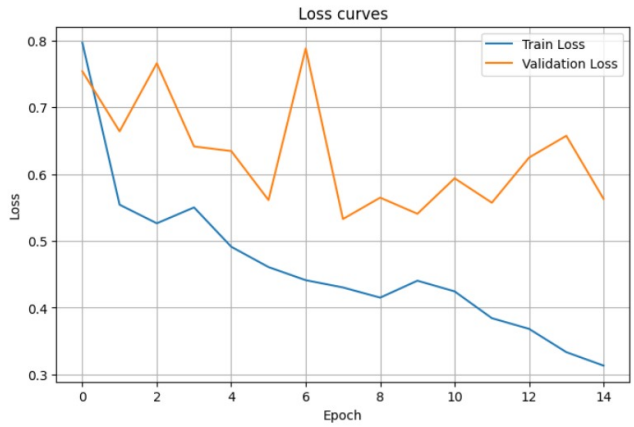

Fig. 3. Loss vs Epoch.

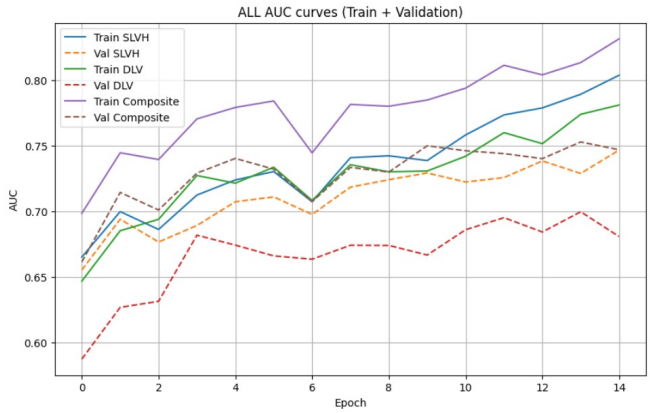

Fig. 4. AUC vs Epoch.

After adjusting the decision thresholds to match a target sensitivity of 66%-70%, as done in the original study, the model effectively identifies positive cases across all tasks. SLVH and DLV show moderate but consistent performance, while the Composite results are the strongest. Composite performs best because it integrates information from both structural abnormalities, making it a more reliable target. Precision is lower for SLVH and DLV, but the higher recall values indicate that the model successfully detects a large proportion of true positive cases. Overall, the classification scores reflect a balanced behavior, with the calibrated thresholds enabling the model to maintain strong sensitivity while preserving reasonable precision.

Task	Precision	Recall	F1-score
SLVH	0.420	0.698	0.524
DLV	0.446	0.712	0.548
Composite	0.718	0.701	0.710

Precision, Recall, and F1-scores for each prediction task.

REFERENCES

[1] P. Elias et al., "CheXchoNet: A chest radiograph dataset with gold-standard echocardiography labels (version 1.0.0)," PhysioNet, 2024. [Online]. Available: https://physionet.org/content/chexchonet/1.0.0/

[2] S. Bhave, V. Rodriguez, T. Poterucha, S. Mutasa, D. Aberle, K. M. Capaccione, Y. Chen, B. Dsouza, S. Dumeer, J. Goldstein, A. Hodes, J. Leb, M. Lungren, M. Miller, D. Monoky, B. Navot, K. Wattamwar, A. Wattamwar, K. Clerkin, D. Ouyang, E. Ashley, V. K. Topkara, M. Maurer, A. J. Einstein, N. Uriel, S. Homma, A. Schwartz, D. Jaramillo, A. J. Perotte, and P. Elias, "Deep learning to detect left ventricular structural abnormalities in chest X-rays," European Heart Journal, vol. 45, no. 22, pp. 2002–2012, Jun. 2024.

[3] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 2017, pp. 2261–2269.

[4] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," CoRR, vol. abs/1412.6980, 2014. Available: https://api.semanticscholar.org/CorpusID:6628106

[5] Z. Liu, Q. Ma, T. Zhang, S. Zhao, X. Gao, T. Sun, and Y. Dai, "Quantitative modeling and uncertainty estimation for small-sample LIBS using Gaussian negative log-likelihood and Monte Carlo dropout methods," Optics & Laser Technology, vol. 181, part A, 111720, 2025.